TT-P-31D
July 6, 1973
SUPERSEDING
Fed. Spec. TT-P-31C
March 7, 1962

FEDERAL SPECIFICATION

PAINT, OIL: IRON-OXIDE, READY-MIXED, RED AND BROWN

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers ready-mixed, red and brown, iron-oxide paint (see 6.3).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

TT-D-651 - Drier; Paint, Liquid.
TT-L-215 - Linseed Oil, Raw, (For Use in Organic Coatings).
TT-P-143 - Paint, Varnish, Lacquer, and Related Materials; Packaging, Packing, and Marking of.
TT-V-121 - Varnish, Spar, Water-Resisting.

Federal Standards;

Fed. Test Method Std. No. 141/Gen. - Paint, Varnish, Lacquer, and Related Materials; Methods of Inspection, Sampling, and Testing.
Fed. Std. No. 595 - Colors.

(Activities outside the Federal Government may obtain copies of Federal Specification Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-P-15328 - Primer (Wash), Pretreatment, Blue (Formula No. 117-B for Metals).

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

American Society for Testing and Materials (ASTM) Standards:

D 2088 - Determination of Low Concentrations of Lead in Paint.
D 2206 - Determination of Low Quantities of Mercury in Paint.
G 26 - Operating Light-and Water-Exposure Apparatus (Xenon-Arc Type) for Exposure of Nonmetalic Materials

(Applications for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103).

Air Pollution Control District

Rule 66

(Application for copies should be addressed to the Air Pollution Control District, County of Los Angeles, CA 90001).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies).

3. REQUIREMENTS

3.1 Pigments. The pigment shall consist of finely ground iron oxide, zinc oxide, and siliceous materials, with or without the addition of a small amount of black pigment to produce the desired color.

3.1.1 Composition. The pigment composition shall be as specified in table I (see 6.4).

TABLE I. Pigment composition

Ingredients	Percent by weight	
	Minimum	Maximum
Total iron oxide (calculated as Fe_2O_3)	60	-
Zinc oxide	12	15
Total calcium pigment (calculated as CaO), in any form soluble in acid	-	1.0
Total ferric oxide (Fe_2O_3), zinc oxide, insoluble siliceous material, and loss on ignition	90	-
Organic coloring matter (other than carbon pigments)	-	None

The paint shall not contain lead compounds in excess of one-half of 1 percent by weight of total nonvolatile, calculated as lead metal. The paint shall not have any added mercury compounds and its mercury metal content shall not exceed a trace quantity (see 6.10).

3.2 Vehicle. The vehicle shall consist of a mixture of 75 percent linseed oil, 15 percent nonreactive spar varnish and 10 percent combined drier and thinner by weight. The spar varnish shall conform to TT-V-121, the linseed oil shall conform to TT-L-215, and the drier shall conform to TT-D-651, type II.

The thinner shall be turpentine, volatile mineral spirits, or a mixture thereof. Where a paint conforming to Air Pollution regulations is required, the supplier shall use "Rule 66" solvents (see 6.5), provided the paint conforms to all other requirements of this specification. The nonvolatile matter in the vehicle shall be not less than 82 percent by weight. The vehicle separated from the ready-mixed paint by means of a supercentrifuge shall conform to the following requirements.

3.2.1 Kauri -eduction. Kauri reduction shall be not less than 200 percent based on the weight of nonvolatile matter in the vehicle.

3.2.2 Cold water resistance (18 hours). When tested as specified in 4.3.1.2, the film shall show no whitening and not more than slight dulling.

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3.3 Quantitative requirements. The iron oxide paint shall meet the quantitative requirements specified in table II.

TABLE II. Quantitative requirements of iron oxide paint

Characteristics	Requirements	
	Minimum	Maximum
Pigment, percent by weight of paint	58	-
Nonvolatile vehicle, percent by weight of vehicle	82	-
Water, percent by weight of paint	-	1.0
Coarse particles and skins (residue retained on No. 325 standard sieve), percent by weight of pigment	-	3
Consistency, Krebs-Stormer, shearing rate 200 rpm:		
Grams	150	250
Equivalent K. U.	72	89
Time to set-to-touch, hours	-	6
Dry through hours	-	18
Weight per gallon, pounds	13.5	-
Specular gloss (without correction for diffuse reflectance)	70	-
Fineness of grind	4	-
Lead content (Nonvolatile basis) percent	-	0.5%

3.4 Qualitative requirements.

3.4.1 Condition in container. The paint shall be thoroughly mixed and ground. The paint shall not settle, cake, or thicken to such a degree that it cannot be redispersed easily with a paddle to a smooth, uniform paint of good brushing consistency.

3.4.2 Color. Unless otherwise specified (see 6.2). The color shall match color number 10076 of Fed. Std. No. 595.

3.4.3 Brushing properties. The paint as received shall brush easily and shall show no brush marks, running, streaking, or sagging.

3.4.4 Appearance. The paint shall dry to a uniform finish free from roughness, grit, and other suface imperfections.

3.4.5 Salt fog. When tested as specified in 4.3.1.1, test panels shall show no rusting, blistering, pitting, or suface defects.

3.4.6 Fungus resistance. When tested as specified in 4.3.1.1, the paint shall not support growth of fungus.

3.4.7 Accelerated weathering. When tested as specified in 4.3.1.1, test panels shall show no blistering, peeling, or cracking, and only slight chalking and discoloration.

3.4.8 Mercury content. When tested as specified in 4.3.1.6, the mercury content of the paint shall not exceed a trace quantity (see 6.10).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.1.1 Material inspection. The supplier is responsible for insuring that materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

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4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.4).

4.3 Quality conformance inspection.

4.3.1 Sampling for testing. Sampling of paint for testing as specified herein shall be in accordance with Fed. Test Method Std. No. 141, method 1031.

4.3.1.1 Physical and chemical properties. The physical and chemical properties of the paint shall be tested in accordance with table III. Non-conformance to a requirement listed in table III for any applicable test shall be cause for rejection of the lot.

TABLE III. Physical and Chemical Properties

Test	Test paragraph Test method of FED. TEST METHOD STD. No. 141	Requirements Paragraphs and tables
Accelerated weathering	ASTM G26, type BH and 4.3.1.4	3.4.7
Condition in container	3011	3.4.1
Percentage of pigment	4021	Table II
Nonvolatile matter in vehicle	4041	Table II
Drying time	4061 1/	Table II
Water	4081	Table II
Coarse particles and skins	4091	Table II
Kauri reduction of supercentrifuged vehicle	4151	3.2.1
Weight per gallon	4184	Table II
Color	4250 2/	3.4.2
Consistency	4281	Table II
Specular gloss	6101 2/	Table II
Fineness of grind	4411 3/	Table II
Pigment analysis	7291	Table I
Salt fog	6061 4/	3.4.5
Fungus resistance	6271 5/	3.4.6
Lead content	4.3.1.5	Table II

1/ Apply the film to a clean, smooth, plane, plateglass panel with a 0.0015-inch (approximately 0.003-inch gap clearance). Bird film applicator or with any other doctor blade which produces a film of the same thickness as that produced by the Bird blade.

2/ Apply the film to a clean, smooth, plane, platelgass panel with a 0.002-inch (approximately 0.004-inch gap clearance) Bird film applicator or any other doctor blade which produces a film of the same thickness as that produced by the Bird blade and allow to air-dry for 48 hours.

3/ Loss on ignition: Ignite 1 gram of the pigment to constant weight. It is safest to use a porcelain crucible for this purpose.

4/ Apply 1.5 plus or minus 0.25 mils of paint to wash-primed (MIL-P-15328 at at 0.3 to 0.5 mil) steel panel expose 300 hours in 5 percent salt fog. Examine for rusting, pitting, blistering, and other surface defects.

5/ FED TEST METHOD STD. No. 141, method 6271, except that (1) Aspergillus oryzae, ATCC 10196 or QM 1273 shall be used as the test organism, (2) test specimens shall be leached before inoculation, (3) the incubation period shall be 7 days, and (4) the sterile distilled water used in the preparation of the inoculum shall contain approximately 0.005 percent of a nontoxic wetting agent (see 6.6).

4.3.1.2 Cold water resistance of supercentrifuged vehicle. Flow the vehicle on a clean tin panel and set the panel in a nearly vertical position to air-dry at room temperature for 72 hours. Place the panel in a beaker containing about 2.5 inches of distilled water at room temperature (immersing the end of panel which was uppermost during the drying period) and leave in water for 18 hours. Remove the panel from the water, dry for 4 hours and examine the film for whitening and dulling. Evidence of whitening or more than slight dulling shall constitute failure of this test.

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4.3.1.3 Brushing properties and appearance. Brush the mixed paint uniformly with a 2- or 2-1/2-inch brush on a 2-foot by 2-foot clean, cold-rolled steel or aluminum panel at a spreading rate of 700 square feet per gallon (21.6 milliliters for an area of 4 square feet). Allow to dry overnight in a nearly vertical position and examine. Evidence of running, streaking, sagging, roughness, grit, or other surface imperfections shall constitute failure of this test.

4.3.1.4 Accelerated weathering. Solvent clean three 3-inch by 9-inch 1020 cold rolled steel panels, and then apply wash primer conforming to MIL-P-15328 at a thickness of 0.3 to 0.5 mil. Allow panels to dry a minimum of 30 minutes, and then apply test paint at a thickness of 1.5 plus or minus 0.25 mils dry film. Allow panels to age 2 weeks in a controlled environment (77°-2°F. relative humidity 60 percent ± 5 percent). Place in xenon weatherometer and test in accordance with ASTM G 26, method BH. Nonconformance to 3.4.7 shall constitute failure of this test.

4.3.1.5 Lead content. The lead content of the paint shall be run in accordance with ASTM D 2088 except that the results shall be reported as percent metal of the nonvolatile portion. A value greater than 0.5 percent shall constitute failure of this test (see 6.11).

4.3.1.6 Mercury content. The analysis for mercury shall be conducted in accordance with ASTM D 2206. The presence of mercury in excess of a trace quantity in the paint shall constitute failure of this test.

4.4 Inspection of preparation for delivery. The packaging, packing, and marking shall be examined to determine compliance with TT-P-143.

5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. The paint shall be packaged, packed, and marked in accordance with TT-P-143. The level of packaging shall be level A, B, or C, and the level of packing shall be level A, B, or C as specified (see 6.2). The paint shall be furnished in the size of container specified (see 6.2).

6. NOTES

6.1 Intended use. This iron-oxide paint is intended for exterior use as a topcoat on wood and metal. It is widely used on tin roofs and is the type frequently referred to as "roof and barn" paint.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in the procurement documents:

(a) Title, number, and date of this specification.
(b) Color required (see 3.4.2).
(c) Level of packaging and level of packing required (see 5.1).
(d) Size of container required (see 5.1).

6.3 The color of the paint under this specification is that of red or brown iron oxide to which has been added a small amount of zinc oxide. Suggested colors for reference in Fed. Std. No. 595 are numbers 10049 and 10076, although the exact color match for paint of this type is not normally required.

6.4 A paint meeting this specification and formulated as indicated below will have the following approximate volume and weight characteristics on a gallon basis:

	Gallon	Pounds
Pigment (58 percent by weight):		
88 percent mineral iron oxide (70 percent Fe_2O_3)	0.2082	7.13
12 percent zinc oxide (American process)	0.0208	0.97
Vehicle (42 percent by weight):		
75 percent linseed oil (Fed. Spec. TT-L-215)	0.5671	4.40
15 percent spar varnish (Fed. Spec. TT-V-121)	0.1164	0.88
10 percent thinner and drier	0.0875	0.58
Total paint	1.0000	13.96

6.5 When solvents conforming to "Rule 66" are required the contracting officer should require the supplier to furnish a certificate of compliance.

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6.6 Cultures of the organism used in the fungus resistance test specified in table III may be obtained from the American Type Culture Collection, 12301 Parklawn Drive, Rockville, Maryland 20852, or for service use, from the Mycology Group, Pioneering Research Laboratories, U. S. Army Natick Laboratories, Natick, MA 01760.

6.7 Iron oxide paint covered by this specification should be purchased by volume (231 cubic inches at 60°F. equals 1 U. S. gallon).

6.8 It is believed that this specification adequately describes the characteristics necessary to secure the desired material and that normally no samples will be necessary prior to award to determine compliance with this specification. If samples with bids are necessary for any particular purpose, they should be specifically asked for in the invitation for bids and the purpose to be served by the bid sample should be definitely stated, the specification to apply in all other respects.

6.9 Trace quantity. A trace quantity is defined as a quantity less than 0.0005 percent.

6.10 Determination of lead content. The following formula will provide the percentage of total nonvolatile by weight:

$$\text{Lead, percent by weight of total nonvolatile} = \frac{A \times 0.86623 \times B}{C \times W} \times 100$$

Where A = Grams of lead oxide (Pb_O2) in ash.
B = Percent of ash in paint divided by 100.
C = Percent of total nonvolatile divided by 100.
W = Grams of ash used.

MILITARY COORDINATING ACTIVITIES

Army, ME
Air Force, 84

Review Activity

Army - MR

User activities:

Army - GL
Navy - MC, YD, SH

Preparing activity:

Army -MC

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